Jitesh Jain

Ph.D. Candidate, School of Interactive Computing, Georgia Tech

jitesh.jj2@gmail.com praeclarumjj3.github.io praeclarumjj3
jitesh-jain-1451b9192 @praeclarumjj

Education

2023 – 2027 expected — **Georgia Tech** — Atlanta, GA
Ph.D. Candidate in Computer Science
Advisor: Prof. Humphrey Shi; CGPA: 4.0/4.0

2019 – 2023 — **IIT Roorkee** — Roorkee, India
B.Tech. in Computer Science and Engineering
CGPA: 9.0/10

Employment

Jan 2026 – Present — **NVIDIA**, High Performance AI Intern w/ Prof. Humphrey Shi
- Leading recipe and infrastructure development for training large (>500B) MoE-based agentic CUDA kernel optimization models.

May 2025 – Dec 2025 — **PRIOR @ Allen AI**, Research Intern w/ Dr. Christopher Clark
- Led development of SAGE across system design, data collection, model training, and evaluation; published at CVPR 2026.

May 2024 – Dec 2024 — **Microsoft Research**, Research Intern w/ Dr. Jianwei Yang
- Led development of VisPer-LM, a visual embedding distillation framework for spatial reasoning in MLLMs; published at NeurIPS 2025.

Jun 2021 – Sep 2022 — **Picsart AI Research**, Research Intern w/ Prof. Humphrey Shi
- Led OneFormer, a train-once universal segmentation framework unifying semantic, instance, and panoptic tasks; published at CVPR 2023.

Selected Publications

For the full publication list, see **Google Scholar**.

Lead-Author Papers

CVPR 2026 — **SAGE: Training Smart Any-Horizon Agents for Long Video Reasoning with Reinforcement Learning** link
Jitesh Jain, Jialuo Li, Zixian Ma, Jieyu Zhang, Chris Dongjoo Kim, Sangho Lee, Rohun Tripathi, Tanmay Gupta, Christopher Clark, Humphrey Shi

NeurIPS 2025 — **Elevating Visual Perception in Multimodal LLMs with Visual Embedding Distillation** link
Jitesh Jain, Zhengyuan Yang, Humphrey Shi, Jianfeng Gao, Jianwei Yang

LAW @ NeurIPS 2025 — **AUGUSTUS: An LLM-Driven Multimodal Agent System with Contextualized User Memory** link
Jitesh Jain*, Shubham Maheshwari*, Ning Yu, Wen-mei Hwu, Humphrey Shi

CVPR 2024 — **VCoder: Versatile Vision Encoders for Multimodal Large Language Models** link
Jitesh Jain, Jianwei Yang, Humphrey Shi

Selected Publications (continued)

CVPR 2023 — **OneFormer: One Transformer to Rule Universal Image Segmentation** link
Jitesh Jain, Jiachen Li*, MangTik Chiu*, Ali Hassani, Nikita Orlov, Humphrey Shi

NIVT @ ICCV 2023 — **SeMask: Semantically Masked Transformers for Semantic Segmentation** link
Jitesh Jain, Anukriti Singh, Nikita Orlov, Zilong Huang, Jiachen Li, Steven Walton, Humphrey Shi

WACV 2023 — **Keys to Better Image Inpainting: Structure and Texture Go Hand in Hand** link
Jitesh Jain*, Yuqian Zhou*, Ning Yu, Humphrey Shi

Co-Authored Papers

CVPR 2026 — **Molmo2: Open Weights and Data for Vision-Language Models with Video Understanding and Grounding** link
Christopher Clark, Jieyu Zhang, Zixian Ma, Jae Sung Park, Mohammadreza Salehi, Rohun Tripathi, Sangho Lee, Zhongzheng Ren, Chris Dongjoo Kim, Yinuo Yang, Vincent Shao, Yue Yang, Weikai Huang, Ziqi Gao, Taira Anderson, Jianrui Zhang, **Jitesh Jain**, George Stoica, Winson Han, Ali Farhadi, Ranjay Krishna

TPAMI under review — **Person Recognition at Altitude and Range: Fusion of Face, Body Shape, and Gait** link
Feng Liu, Nicholas Chimitt, Lanqing Guo, **Jitesh Jain**, Aditya Kane, Minchul Kim, Wes Robbins, Yiyang Su, Dingjiang Ye, Xingguang Zhang, Jie Zhu, Siddharth Satyakam, Christopher Perry, Stanley H. Chan, Arun Ross, Humphrey Shi, Zhangyang Wang, Anil Jain, Xiaoming Liu

NeurIPS 2024 — **CuMo: Scaling Multimodal LLM with Co-Upcycled Mixture-of-Experts** link
Jiachen Li, Xinyao Wang, Sijie Zhu, Chia-Wen Kuo, Lu Xu, Fan Chen, **Jitesh Jain**, Humphrey Shi, Longyin Wen

ECCV 2024 — **Benchmarking Object Detectors with COCO: A New Path Forward** link
Shweta Singh*, Aayan Yadav*, **Jitesh Jain**, Humphrey Shi, Justin Johnson, Karan Desai

Awards & Honors

2026 — Selected for ACM AI Leadership Summit Future Leaders Doctoral Consortium
Selected for CVPR 2026 Doctoral Consortium
2025 — NeurIPS Scholar Award
2019 — All India Rank 252, JEE Advanced
All India Rank 454, JEE Main

Service & Talks

Reviewer — NeurIPS 2026; CVPR 2026; NeurIPS 2025; ICCV 2025; ICML 2025; CVPR 2025; ECCV 2024; CVPR 2024; WACV 2023.
Organizer — CODA AI Synapse, Georgia Tech, 2023–2025.
Talks — Invited Talk at Isola Lab@MIT, 2026.